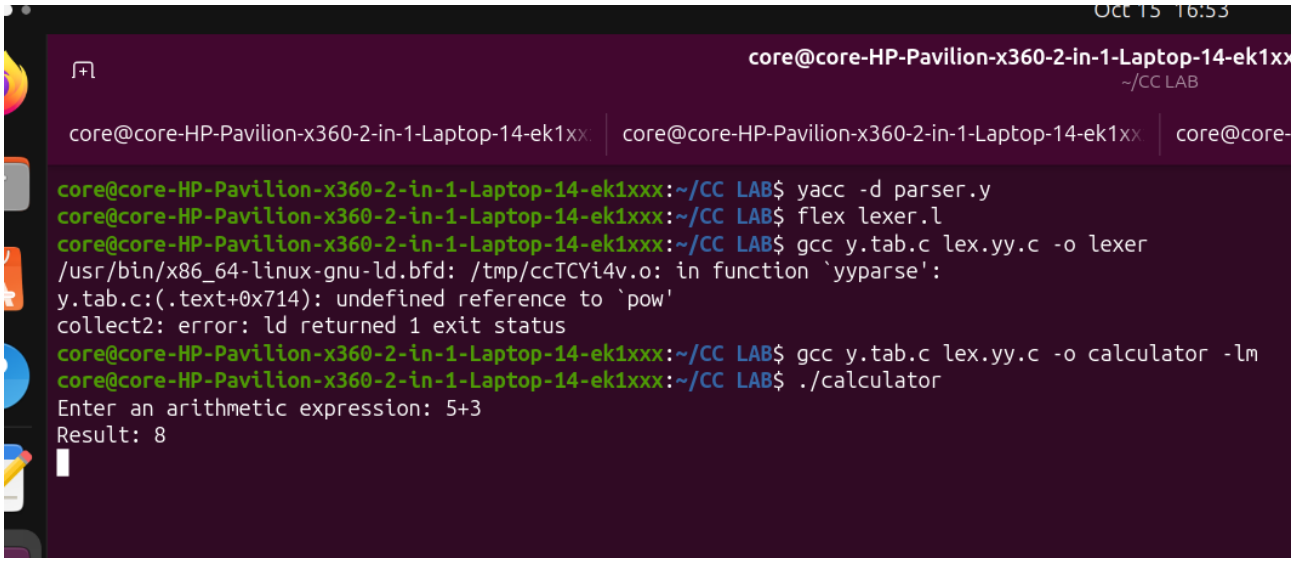
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Batch : B2



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Open

parser.y

```
%{
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int yylex(void);
int yyerror(const char *s);
%}

%union {
    int num;
}

%token <num> NUMBER
%type <num> expr

%left '+' '-'
%left '*' '/' '%'
%right '^'
%right UMINUS

%%

input:
      expr '\n' {
          printf("Result: %d\n", $1);
      }
    ;

expr:
      NUMBER {
          $$ = $1;
      }
    | expr '+' expr {
          $$ = $1 + $3;
      }
    | expr '-' expr {
          $$ = $1 - $3;
      }
```

Open

parser.y

```
    | expr '/' expr {
          if ($3 == 0) {
              printf("Error: Division by zero\n");
              YYABORT;
          }
          $$ = $1 / $3;
      }
    | expr '%' expr {
          if ($3 == 0) {
              printf("Error: Modulo by zero\n");
              YYABORT;
          }
          $$ = $1 % $3;
      }
    | expr '^' expr {
          $$ = (int)pow($1, $3);
      }
    | '(' expr ')' {
          $$ = $2;
      }
    | '-' expr %prec UMINUS {
          $$ = -$2;
      }
    ;

%%

int yyerror(const char *s)
{
    printf("Invalid expression\n");
    return 0;
}

int main(void)
{
    printf("Enter an arithmetic expression: ");
    yyparse();
    return 0;
}
```